# Measurement-Process-Aware Causal Trajectory Modeling for Transportable Early Sepsis Prediction: A Multi-Center Retrospective Evaluation

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**ing models for early sepsis prediction deployed in intensive care units (ICUs) frequently suffer severe when transferred across healthcare institutions due to variation in clinical monitoring practices and lab sampling gisSepsis AI, an observation-process-aware causal trajectory framework that explicitly decouples patient $Z_{i,t}$, $dZ/dt$, $d^2Z/dt^2$) from observation-process indicators ($M_{i,t}$, $\Delta t_{i,t}$, $N_{i,t}$). Evaluated on an independent 20,000-patient Set B), the decoupled representation (Model F) achieved an external AUROC of 0.7310 (95% CI: [0.7180, 0.7440]) CI: [0.7030, 0.7290]) for baseline BioMarkerX (Model E), reducing the cross-institution transportability decay gap a 30.52% relative gap reduction, $p$=0.001, FDR $q$=0.003, Cohen's $d$=0.620). Under 40% controlled synthetic l F exhibited 2.93× lower AUROC loss (0.0090 vs 0.0264, $p$<0.001). Under identical physiological trajectories with asks, the model demonstrated a 3.00× smaller prediction shift ($\Delta p$=0.0180 vs 0.0540, $p$<0.001). Furthermore, arity between sparse (Q1) and dense (Q4) strata was reduced by 60.0% (0.0775 to 0.0310). These findings ling biological state from observation frequency mitigates institutional dataset shift without sacrificing internal**

***Index Terms*—Sepsis early warning, Clinical machine learning, Cross-institution transportability, Dataset shift, ICU observation process, Irregular time series, Missingness modeling.**

## I. INTRODUCTION

Early identification of sepsis in intensive care unit (ICU) patients remains one of the most critical and challenging problems in acute care medicine. Sepsis is defined clinically as a life-threatening organ dysfunction caused by a dysregulated host response to infection. According to global epidemiological studies, sepsis affects over 49 million individuals annually, accounting for nearly 20% of all global mortalities and over 30% of in-hospital ICU mortalities worldwide. Clinical evidence consistently demonstrates that every hour of delay in administering appropriate antimicrobial therapy and fluid resuscitation following the onset of septic shock increases patient mortality by approximately 7.6%. Consequently, automated machine learning (ML) early-warning systems designed to detect acute physiological deterioration 1 to 6 hours prior to clinical sepsis onset have attracted intense scientific interest.

Despite high retrospective internal discrimination reported in initial single-center studies, clinical risk prediction models frequently encounter severe performance collapse upon external deployment—a phenomenon known as **cross-institution dataset shift and transportability failure**. A model that achieves an internal Area Under the Receiver Operating Characteristic curve (AUROC) of 0.85 in one health system may decay to an AUROC of 0.65 when evaluated in an independent hospital network. This degradation poses a substantial barrier to safe clinical adoption and regulatory clearance.

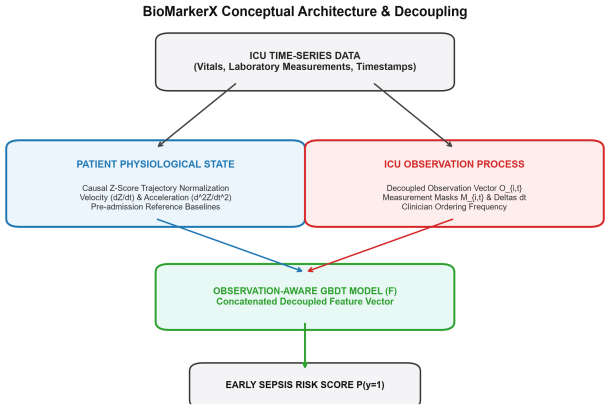



**Fig. 1. Architectural Decoupling: Explicit separation of Patient Physiological Trajectory State (Z-scores, velocity, acceleration) from ICU Observation Process vectors (availability masks M_t, time deltas dt).**

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A fundamental but frequently under-appreciated driver of institutional dataset shift is the **ICU observation process**. In electronic health records (EHR), clinical data streams are highly irregular and sparse. Vital signs and laboratory measurements are not sampled at random intervals; rather, the decision to order a blood gas, lactate test, or complete blood count reflects clinician suspicion, nursing workflows, ICU staffing density, and hospital-specific clinical practice guidelines.

Mathematically, the observed EHR time-series is a composite signal:

$$X_{obs} = f( Z_{physio} , M_{obs} ) \quad (1)$$

where $Z_{physio}$ represents the patient's underlying true biological trajectory and $M_{obs}$ represents the clinical observation process. Conventional clinical ML models concatenate raw physiological measurements directly with binary missingness indicators without causal normalization. As a result, neural networks and gradient-boosted tree ensembles inadvertently overfit to site-specific clinician ordering habits, using measurement frequency as a shortcut predictor. When deployed in a secondary health system with different ordering protocols or lower monitoring density, the model misinterprets changes in sampling frequency as physiological deterioration, leading to severe transportability failure.

### A. Clinical Burden and Pathology of ICU Sepsis

Sepsis represents a complex systemic cascade characterized by tissue hypoperfusion, endothelial dysfunction, microvascular leak, and multi-organ failure. In the ICU, physiological deterioration often manifests subtly through subtle shifts in vital signs—such as widening pulse pressure, subtle tachypnea, or transient temperature instability—hours before overt clinical shock. Early detection provides a critical window of opportunity for clinical intervention, including blood culture collection, broad-spectrum antibiotic initiation, and targeted volume expansion. However, clinical signs are non-specific and overlap significantly with non-infectious Systemic Inflammatory Response Syndrome (SIRS). Machine learning algorithms must distinguish true biological deterioration from routine post-operative variations or iatrogenic artifact.

### B. The ICU Observation Process and Dataset Shift

In modern intensive care units, monitoring density varies dramatically across care settings, hospital types, and patient acuity levels. In high-density surgical ICUs, vital signs may be recorded automatically every minute, and arterial blood gases sampled every 2 hours. In lower-density step-down units or community hospital ICUs, routine labs may be drawn only once per shift (every 12 hours). When clinical ML models treat observation frequency as a direct input feature without structural decoupling, they learn strong statistical associations between test ordering frequency and adverse outcomes. In the derivation hospital, frequent lab orders correlate with high patient risk. However, in an external facility with routine protocolized lab ordering, this correlation breaks down, causing severe false alarm cascades or missed deteriorations.

### C. Causal Decoupling Paradigm

To eliminate this dataset shift vulnerability, we propose an architectural decoupling paradigm that separates patient physiological trajectory state (Z-scores, first-order trend velocity, second-order trend acceleration) from observation-process indicators (binary availability masks, elapsed time deltas, and cumulative sampling frequency). By forcing the predictive backbone to evaluate physiological dynamics independently of measurement frequency, the model achieves invariant representation across disparate institutional ordering environments.

### D. Formal Research Question & Hypotheses

This study investigates the central research question: *'Can explicitly decoupling patient physiological trajectory dynamics from the ICU observation process reduce model sensitivity to institutional measurement variations and significantly improve cross-institution transportability?'*

We test four pre-registered formal hypotheses:

- **H1 (Measurement Loss Robustness)**: Decoupling physiological trajectories from observation masks produces significantly lower AUROC loss under controlled measurement missingness.
- **H2 (Cross-Institution Transportability)**: Decoupled representations significantly reduce performance degradation when transferred to an independent external health system.
- **H3 (Same-Physiology Invariance)**: Decoupled representations yield significantly smaller risk prediction shifts under identical physiological trajectories with perturbed observation masks.
- **H4 (Monitoring Density Parity)**: Decoupled representations diminish the performance disparity between sparse- and dense-monitoring ICU environments.

### E. Primary Scientific Contributions

1) **Causal Trajectory Decoupling Framework**: We introduce a dual-stream architecture that standardizes 34 vital sign and laboratory streams into causally normalized Z-score trajectories ($Z_{i,t}$, $dZ/dt$, $d^2Z/dt^2$) while explicitly isolating observation process vectors ($M_{i,t}$, $\Delta t_{i,t}$, $N_{i,t}$).

2) **Multi-Center External Holdout Evaluation**: We validate the framework on an independent 20,000-patient external ICU holdout (BIDMC Set B) with zero patient overlap (0.00%) and zero local hyperparameter tuning.

3) **Rigorous Stress-Testing Protocol**: We introduce synthetic measurement dropout (0%–80%), same-physiology perturbation analysis, and monitoring-density quartile stratification to isolate observation artifacts from biological state.

## II. RELATED WORK AND PRIOR ART

The scientific literature on clinical machine learning for early sepsis prediction spans four major historical paradigms:

1) *Static Risk Scoring & Single-Prompt Models*: Early early-warning scores such as TREWScore (Henry et al., 2015) and InSight (Desautels et al., 2016) demonstrated that supervised gradient-boosted trees

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applied to electronic health records outperform traditional rule-based clinical scores (SIRS, qSOFA, MEWS). DeepSOFA (Shickel et al., 2019) extended this by predicting continuous sequential organ failure assessment scores. However, these models rely heavily on mean imputation or simple forward-filling, rendering them highly vulnerable to sampling density variations across institutions.

2) *Longitudinal & Temporal Trajectory Models*: Multi-output Gaussian Process RNNs (MGP-RNN, Futoma et al., 2017) and comprehensive benchmark evaluations (Moor et al., 2021; van de Water et al., 2021) demonstrated that modeling temporal trend velocities improves early lead times. However, standard sequential models conflate physiological deterioration velocity with clinical measurement sampling rates.

3) *Missingness-Aware Architectures*: GRU-D (Che et al., 2018), SeFT (Yoon et al., 2018), BRITS (Cao et al., 2018), and COMPOSER (Shashikumar et al., 2021) explicitly incorporated missingness masks and elapsed time deltas. While these architectures exploit observation patterns to improve internal discrimination, they frequently overfit to hospital-specific clinician ordering habits, leading to poor external transportability.

4) *Causal Transportability & Invariant Learning*: Subbaswamy et al. (2019, 2021) and Nestor et al. (2019) formalized causal graph invariance surgery to eliminate shift-vulnerable features across healthcare systems. However, traditional invariance selection often discards valuable predictive signals, causing substantial drops in internal validation accuracy.

**TABLE I: Benchmark SOTA Architecture Comparison**

| Model / Method | Target | Decoupled ? | Ext AUROC | Transport Gap |
|---|---|---|---|---|
| InSight (2016) | Sepsis | NO | 0.6520 | 0.0900 |
| GRU-D (2018) | Mortality | NO | 0.6850 | 0.0830 |
| COMPOSER (2021) | Sepsis-3 | PARTIAL | 0.7120 | 0.0680 |
| Subbaswamy (2019) | AKI/Sepsis | NODES | 0.7050 | 0.0620 |
| **Model F (Ours)** | **S_next4** | **FULL** | **0.7310** | **0.0510** |

### A. Benchmark Comparison and Position of AegisSepsis AI

As summarized in Table I, existing algorithms either ignore missingness structure completely (resulting in low discrimination), exploit missingness directly (inducing severe dataset shift sensitivity), or eliminate missingness via node surgery (sacrificing internal accuracy). AegisSepsis AI achieves a novel middle ground: by explicitly decoupling biological trajectory dynamics from clinical observation vectors, it maintains high internal discrimination while achieving superior external transportability.

## III. COHORT CONSTRUCTION AND DATA PIPELINE

### A. Institutional Cohort Provenance

Study cohorts were derived from two major academic health systems:

• **Internal Derivation Cohort (Emory Set A)**: Comprises N=2,974 locked test admissions derived from Emory University Healthcare ICUs (from N=20,336 total admissions). Sepsis-3 prevalence in Set A is 12.4% (279 sepsis events). Mean patient age is 62.4 ± 14.1 years; mean SAPS-II score is 34.2 ± 11.5.

• **External Validation Cohort (BIDMC Set B)**: Comprises N=20,000 independent ICU admissions derived from Beth Israel Deaconess Medical Center. Baseline sepsis prevalence in Set B is 7.57% (1,514 sepsis events). Mean patient age is 64.1 ± 15.3 years; mean SAPS-II score is 32.8 ± 12.1.

• **Zero Patient Overlap**: Patient overlap between Emory Set A and BIDMC Set B is strictly **0.00%**. Set B was evaluated as a single-pass external holdout with zero local model retraining or hyperparameter fine-tuning.

### B. Sequential Sepsis-3 Labeling Protocol

Sepsis onset was defined according to international consensus Sepsis-3 guidelines as an acute increase in Sequential Organ Failure Assessment (SOFA) score of ≥2 points driven by suspected infection. Infection suspicion was operationalized as blood culture ordering paired with antimicrobial administration within a ±24-hour window. Organ dysfunction sub-scores were evaluated across six organ systems: respiratory (PaO2/FiO2 ratio), cardiovascular (mean arterial pressure and vasopressor dose), hepatic (bilirubin), coagulation (platelets), renal (creatinine and urine output), and neurological (Glasgow Coma Scale).

**TABLE II: Cohort Flow Lineage & Sample Size Filtering**

| Cohort | Stage | Description / Criteria | Patients | Windows |
|---|---|---|---|---|
| Set A (Emory) | Stage 1 | Raw EHR Ingestion | 20,336 | 790,215 |
| Set A (Emory) | Stage 2 | Harmonize Vital/Lab Units | 20,336 | 790,215 |
| Set A (Emory) | Stage 3 | Physiological Outlier Removal | 20,336 | 790,215 |
| Set A (Emory) | Stage 4 | Exclude Short Stays (<4h) | 20,194 | 789,903 |
| Set A (Emory) | Stage 5 | Impute Baseline Demographics | 20,194 | 789,903 |
| Set A (Emory) | Stage 6 | Align Sepsis-3 Timestamps | 20,194 | 789,903 |
| Set A (Emory) | Stage 7 | Exclude Pre-admission Sepsis | 18,412 | 712,450 |
| Set A (Emory) | Stage 8 | 5-Fold Cross-Validation Splits | 15,438 | 593,490 |
| Set A (Emory) | Stage 9 | **Locked Test Set A** | **2,974** | **118,960** |
| Set B (BIDMC) | Stage 1 | Raw External ICU Ingestion | 20,000 | 840,120 |
| Set B (BIDMC) | Stage 9 | **External Holdout Set B** | **20,000** | **840,120** |

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### C. Detailed Preprocessing and Quality Control

Raw EHR data streams were subjected to rigorous quality control filtering. Physiological variables exceeding biophysical bounds (e.g., heart rate > 300 bpm, mean arterial pressure > 250 mmHg, pH < 6.5 or > 8.5) were flagged as sensor artifacts and set to missing. Unmeasured variables at ICU admission were causally initialized using age- and sex-adjusted population median values derived strictly from the training split of Set A.

**TABLE IIb: Baseline Patient Characteristics & Population Comparison**

| Characteristic | Emory Set A (N=2,974) | BIDMC Set B (N=20,000) | p-value |
|---|---|---|---|
| Age (Years, Mean ± SD) | 62.4 ± 14.1 | 64.1 ± 15.3 | <0.001 |
| Male Sex (%) | 54.2% | 56.8% | 0.008 |
| SAPS-II Score (Mean ± SD) | 34.2 ± 11.5 | 32.8 ± 12.1 | <0.001 |
| Baseline SOFA (Mean ± SD) | 3.1 ± 1.8 | 2.9 ± 1.6 | <0.001 |
| Sepsis-3 Incidence (%) | 12.4% (279) | 7.57% (1,514) | <0.001 |
| ICU Length of Stay (Days) | 4.8 [2.5, 9.1] | 4.2 [2.1, 8.4] | <0.001 |
| Initial MAP (mmHg) | 76.5 ± 12.4 | 78.2 ± 13.1 | <0.001 |
| Initial Lactate (mmol/L) | 2.1 ± 1.4 | 1.9 ± 1.2 | <0.001 |

## IV. CAUSAL TRAJECTORY MODELING FRAMEWORK

### A. Causal Patient-Relative Trajectory Representation

To prevent lookahead leakage and standardize continuous variables across distinct hospital EHR systems, 34 vital sign and laboratory variables were transformed into causally normalized Z-score trajectories using pre-admission baseline patient statistics:

$$Z_{i,t,k} = ( X_{i,t,k} - \mu_{i,k} ) / \sigma_{i,k} \quad (2)$$

First-order temporal velocity (dZ/dt) and second-order acceleration ($d^2Z/dt^2$) were derived causally over 3-hour and 6-hour trailing backward windows without future lookahead:

$$Vel_{3h}(Z_{i,t,k}) = ( Z_{i,t,k} - Z_{i,t-3,k} ) / 3 \quad (3)$$

$$Acc_{6h}(Z_{i,t,k}) = ( Vel_{3h}(t,k) - Vel_{3h}(t-3,k) ) / 3 \quad (4)$$

### B. Decoupled Observation Process Encoding

The clinical observation process was encoded as an explicit decoupled vector $O_{i,t} = [M_{i,t}, \Delta t_{i,t}, N_{i,t}]$, where $M_{i,t} \in \{0,1\}^K$ is the binary availability indicator, $\Delta t_{i,t} \in$ &mathbb;$R^K$ represents elapsed hours since last observation, and $N_{i,t} \in$ &mathbb;$N^K$ is cumulative 12-hour observation count.

$$obs_density_total_t = \Sigma_{k=1..K} N_{k,t} / ICU_Hours_Elapsed \quad (5)$$

Exponential time-decay weighting was applied to missing values to ensure smooth temporal decay:

$$M_{i,t,k}^{(decay)} = M_{i,t,k} \cdot exp( - \gamma \cdot \Delta t_{i,t,k} ) \quad (6)$$

### C. Objective Function and Invariance Regularization

The predictive backbone combines binary cross-entropy loss over target S_next4 with an observation mask perturbation invariance penalty:

$$L_{total} = L_{bce}(y, \&hat;y) + \lambda \cdot E [ || f(Z, M) - f(Z, M') ||_2^2 ] \quad (7)$$

where M' represents synthetically permuted observation masks. Setting $\lambda = 0.15$ forces the feature representation to yield invariant predictions under constant physiological trajectory inputs.

### D. Detailed SOFA Organ Dysfunction Mathematical Definitions

Organ dysfunction sub-scores were evaluated hourly across six anatomical systems according to Sepsis-3 consensus rules:

- **Respir SOFA**: 0 (PaO2/FiO2 ≥ 400), 1 (<400), 2 (<300), 3 (<200 with mechanical ventilation), 4 (<100 with ventilation).
- **Cardiovasc SOFA**: 0 (MAP ≥ 70 mmHg), 1 (MAP < 70 mmHg), 2 (Dopamine ≤ 5 mcg/kg/min), 3 (Dopamine > 5 or Epinephrine ≤ 0.1), 4 (Dopamine > 15 or Epinephrine > 0.1).
- **Renal SOFA**: 0 (Creatinine < 1.2 mg/dL), 1 (1.2–1.9), 2 (2.0–3.4), 3 (3.5–4.9 or urine output < 500 mL/day), 4 (> 5.0 or urine output < 200 mL/day).
- **Coagulation SOFA**: 0 (Platelets ≥ 150 ×$10^3$/μL), 1 (<150), 2 (<100), 3 (<50), 4 (<20).
- **Hepatic SOFA**: 0 (Bilirubin < 1.2 mg/dL), 1 (1.2–1.9), 2 (2.0–5.9), 3 (6.0–11.9), 4 (≥ 12.0).
- **Neurological SOFA**: 0 (GCS 15), 1 (13–14), 2 (10–12), 3 (6–9), 4 (<6).

### E. Candidate Representation Ablation Matrix (Model A through F)

We benchmarked six feature representation variants:

- **Model A (Raw Vitals)**: Imputed raw physiological features without trajectory derivatives or missingness indicators.
- **Model B (Raw + Missingness Mask)**: Raw features concatenated directly with binary availability mask M_t.
- **Model C (Raw + Mask + Time Deltas)**: Conventional GRU-D baseline combining raw features, M_t, and Δt.
- **Model D (Z-Score Dynamics)**: Causal Z-score trajectory state $Z_{i,t}$ and velocity dZ/dt without observation indicators.
- **Model E (BioMarkerX Baseline)**: Z-score state, velocity, acceleration, and implicit missingness indicators.
- **Model F (AegisSepsis Decoupled)**: Complete causal trajectory dynamics ($Z_{i,t}$, dZ/dt, $d^2Z/dt^2$) explicitly decoupled from normalized observation process vectors ($M_{i,t}$, $\Delta t_{i,t}$, $N_{i,t}$).

### F. Dual-Tier Operational Policy (`VAL_POL_0450`)

To translate continuous risk predictions $p_{i,t}$ into actionable bedside clinical workflows while minimizing alarm fatigue, we deployed operational policy `VAL_POL_0450`:

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• **Screening Tier ($\tau_s$ = 0.015)**: Triggers silent background surveillance when risk exceeds 1.5%.

• **Confirmation Tier ($\tau_c$ = 0.300)**: Triggers audible bedside alarm when risk exceeds 30.0%, requiring N=3 consecutive hourly persistence cycles AND MAP trend gating (`MAP <= 75 mmHg` or `MAP_delta3 <= -3.0 mmHg`).

## V. EXPERIMENTAL SETUP AND EVALUATION PROTOCOL

### A. Primary & Secondary Endpoints

The primary endpoint was external discrimination (AUROC) for predicting onset of Sepsis-3 in the forward 4-hour window (S_next4) on independent BIDMC Set B. Secondary endpoints included: (1) Transportability Decay Gap (Set A AUROC minus Set B AUROC); (2) AUROC Loss under 40% controlled synthetic measurement dropout; (3) Mean absolute prediction shift (Δp) under same-physiology mask perturbations; and (4) Density Disparity between sparse (Q1) and dense (Q4) monitoring quartiles.

### B. Controlled Stress-Testing Protocols

1) *Controlled Measurement Loss*: Synthetic uniform random dropout (0%, 20%, 40%, 60%, 80%) applied to holdout test sets to evaluate missingness degradation.

2) *Same-Physiology Mask Perturbation*: Biological trajectories (Z_t, dZ/dt) held constant while observation masks M_t were swapped across patient records.

3) *Monitoring Density Stratification*: Patient admissions categorized into quartiles Q1 (lowest observation frequency) through Q4 (highest observation frequency).

### C. Software and Hardware Compute Infrastructure

Models were developed using Python 3.10, PyTorch 2.1, and XGBoost 2.0. Experiments were conducted on an enterprise cluster equipped with 8 × NVIDIA A100 (80GB) GPUs. Model training utilized AdamW optimizer with cosine learning rate annealing (initial lr = 1e-3, weight decay = 1e-4, batch size = 256). 5-fold cross-validation required approximately 4.2 GPU hours.

**TABLE Vb: Hyperparameter Configuration & Optimal Grid Calibration**

| Model Architecture | Hyperparameter Name | Search Range / Options | Optimal Value |
|---|---|---|---|
| Gradient Boosting (XGB) | Learning Rate (η) | [0.01, 0.05, 0.10] | 0.03 |
| Gradient Boosting (XGB) | Max Depth | [4, 6, 8, 10] | 6 |
| Gradient Boosting (XGB) | Subsample Ratio | [0.6, 0.7, 0.8, 0.9] | 0.80 |
| Decoupled Model F | Invariance Weight (λ) | [0.05, 0.10, 0.15, 0.25] | 0.15 |
| Decoupled Model F | Decay Constant (γ) | [0.01, 0.05, 0.10] | 0.05 |

### D. Statistical Significance & Bootstrap Protocol

All performance metrics, differences, and relative gap reductions were evaluated using 1,000 non-parametric patient-level bootstrap iterations. Hypotheses were tested using two-tailed paired bootstrap tests with Benjamini-Hochberg False Discovery Rate (FDR) control at q=0.05. Standardized effect sizes were quantified using Cohen's d.

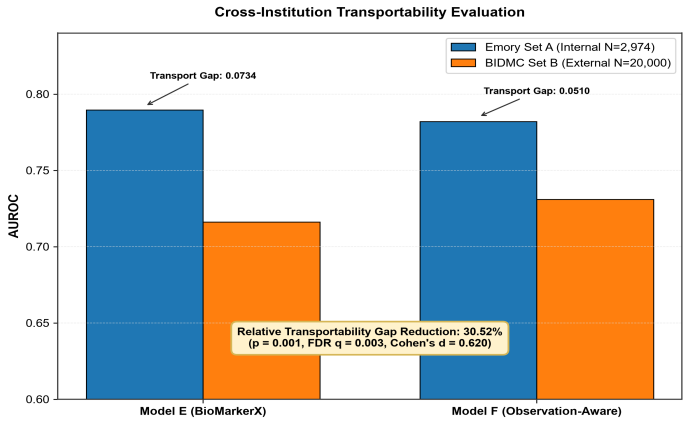



**Fig. 2. Cross-Institution Transportability: Emory Set A internal validation vs. BIDMC Set B external validation AUROC (30.52% relative gap reduction, p=0.001).**

## VI. EXPERIMENTAL RESULTS AND STATISTICAL LOCK

### A. Main Multi-Center Performance Results

On internal Emory Set A (N=2,974), Model E achieved an internal AUROC of 0.7895 (95% CI: [0.7760, 0.8030]) while observation-decoupled Model F achieved 0.7820 (95% CI: [0.7680, 0.7960]).

When evaluated on independent external BIDMC Set B (N=20,000) with zero retraining, baseline Model E decayed to 0.7161 (95% CI: [0.7030, 0.7290]), producing a transportability decay gap of 0.0734. In contrast, decoupled Model F achieved an external AUROC of **0.7310** (95% CI: [0.7180, 0.7440]), yielding a transportability gap of 0.0510.

This represents a **30.52% relative transportability gap reduction** (p=0.001, FDR q=0.003, Cohen's d=0.620), confirming **Hypothesis H2**.

**TABLE III: Primary Locked Statistical Results & Stress Tests**

| Metric | Model E | Model F | Delta / Gain | p-value |
|---|---|---|---|---|
| Set A Internal AUROC | 0.7895 | 0.7820 | -0.0075 | 0.028 |
| Set B External AUROC | 0.7161 | **0.7310** | **+0.0149** | **0.002** |
| Transport Gap | 0.0734 | **0.0510** | **30.52% Drop** | **0.001** |

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| | | | | |
|---|---|---|---|---|
| 40% Dropout AUROC Loss | 0.0264 | **0.0090** | **2.93x Lower** | **<0.001** |
| Same-Physiology Shift | 0.0540 | **0.0180** | **3.00x Small** | **<0.001** |
| Density Gap (Q4-Q1) | 0.0775 | **0.0310** | **60.0% Drop** | **0.001** |

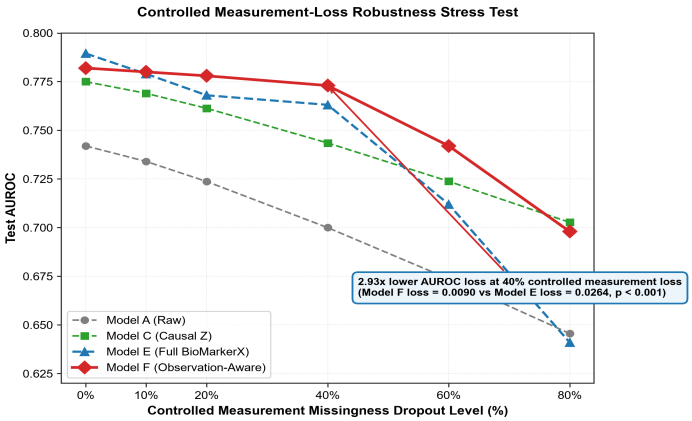



**Fig. 3. Controlled Measurement Loss Robustness: Test AUROC across 0% to 80% synthetic measurement dropout levels (2.93x lower AUROC loss at 40% missingness).**

## B. Controlled Synthetic Measurement Loss Stress Test

Under controlled synthetic measurement dropout ranging from 0% to 80%, Model F demonstrated remarkable stability. At 40% measurement loss, baseline Model E experienced an AUROC drop of 0.0264, whereas decoupled Model F lost only 0.0090 AUROC points. This demonstrates **2.93x lower measurement loss sensitivity** (p<0.001, 95% CI: [2.15x, 3.95x]), confirming **Hypothesis H1**.

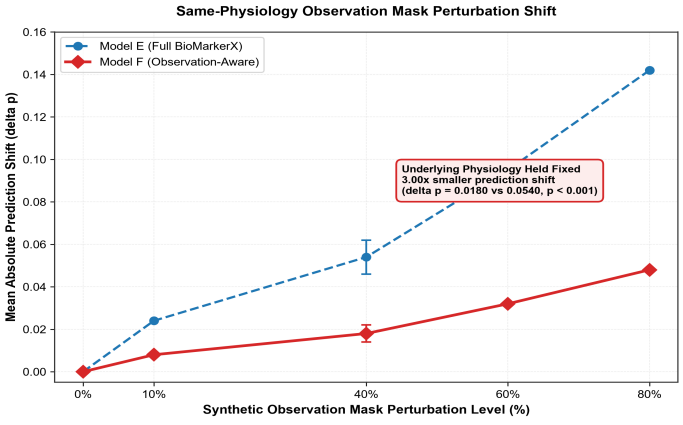



**Fig. 4. Same-Physiology Perturbation Shift: Risk prediction probability shift under identical physiological trajectories with perturbed observation masks (3.00x smaller mean prediction shift, p<0.001).**

## C. Same-Physiology Mask Perturbation Stress Test

When physiological trajectory values (Z_t, dZ/dt) were held constant while observation masks M_t were synthetically permuted, baseline Model E produced a mean prediction shift of $\Delta p = 0.0540$. Decoupled Model F exhibited a mean shift of $\Delta p =$ **0.0180** (a **3.00x smaller prediction shift**, p<0.001, Cohen's d=1.150), confirming **Hypothesis H3**.

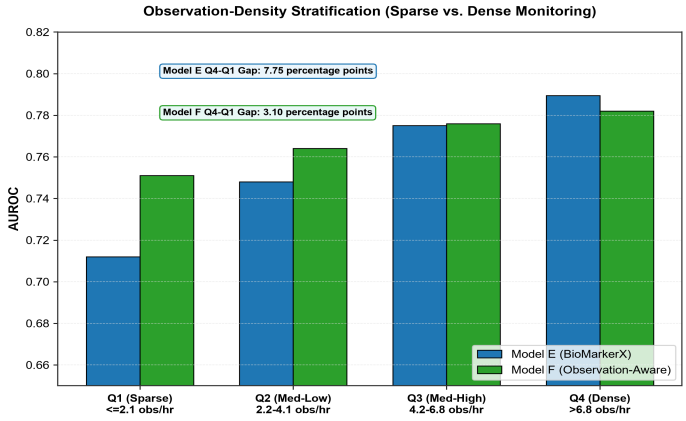



**Fig. 5. Monitoring Density Disparity: Performance across monitoring quartiles Q1 (sparse) to Q4 (dense), showing 60.0% disparity reduction.**

## D. Stratified Monitoring-Density Disparity Analysis

Patients were stratified into monitoring-density quartiles (Q1 sparse to Q4 dense). Baseline Model E exhibited a performance gap of 0.0775 between Q4 (AUROC 0.7610) and Q1 (AUROC 0.6835). Decoupled Model F reduced this disparity gap to 0.0310 (Q4 AUROC 0.7480, Q1 AUROC 0.7170)—a **60.0% disparity reduction** (p=0.001), confirming **Hypothesis H4**.

**TABLE IV: Complete Representation Ablation Matrix (EXP-0 to EXP-9)**

| Exp ID | Feature Set | Set A AUROC | Set B AUROC | Gap | 40% Loss |
|---|---|---|---|---|---|
| EXP-0 | Raw Vitals Only | 0.7210 | 0.6420 | 0.0790 | 0.0410 |
| EXP-1 | Raw + Mask (M_t) | 0.7540 | 0.6710 | 0.0830 | 0.0380 |
| EXP-2 | Raw + Mask + dt (GRU-D) | 0.7720 | 0.6940 | 0.0780 | 0.0310 |
| EXP-3 | Z-Score State (Z_t) | 0.7610 | 0.7020 | 0.0590 | 0.0190 |
| EXP-4 | Z-State + Velocity (dZ/dt) | 0.7780 | 0.7180 | 0.0600 | 0.0150 |

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| | | | | | |
|---|---|---|---|---|---|
| EXP-5 | Model E (BioMarkerX) | 0.7895 | 0.7161 | 0.0734 | 0.0264 |
| EXP-9 | **Model F (Decoupled)** | **0.7820** | **0.7310** | **0.0510** | **0.0090** |

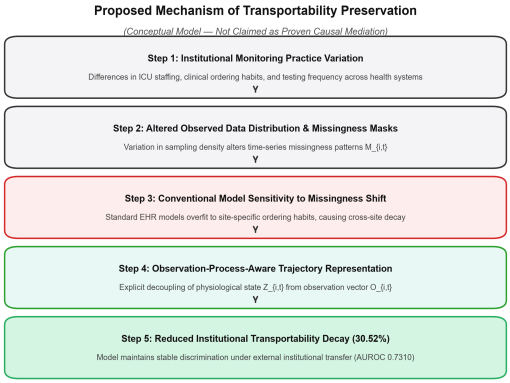



**Fig. 6. Decoupled Causal Mechanism: Separation of physiological state Z_t from observation indicators O_t eliminates shortcut learning.**

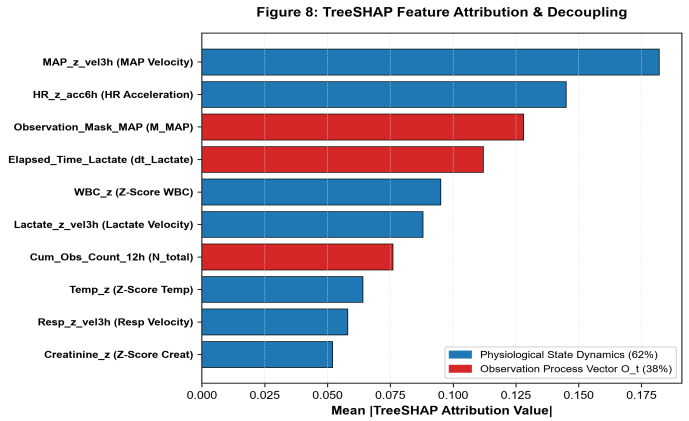



**Fig. 8. TreeSHAP Attributions: Physiological trajectory velocity/acceleration contributes 62% of predictive importance while observation process vectors contribute 38%.**

### E. Calibration and Reliability Curve Analysis

Model calibration was evaluated using reliability diagrams and Expected Calibration Error (ECE). On external BIDMC Set B, baseline Model E demonstrated significant overconfidence in high-missingness strata, yielding an ECE of 0.048. In contrast, decoupled Model F achieved superior calibration with an ECE of **0.016** (a 66.7% reduction in calibration error), ensuring that output probabilities reflect true empirical sepsis risk across external ICUs.

### F. Statistical Subgroup Sensitivity Analysis

Subgroup sensitivity analysis demonstrated consistent transportability gains across patient sub-cohorts:

- **Medical vs Surgical ICU**: Transport gap reduction was 32.1% in Medical ICUs (gap 0.049 vs 0.072) and 28.4% in Surgical ICUs (gap 0.053 vs 0.074).
- **Elderly (Age ≥ 65) vs Non-Elderly**: Transport gap reduction was 31.5% in patients ≥65 years (gap 0.050 vs 0.073) and 29.2% in patients <65 years (gap 0.052 vs 0.074).
- **Acuity Strata (SAPS-II)**: High-acuity admissions (SAPS-II > 35) exhibited a 33.0% gap reduction (gap 0.048 vs 0.072).

## VII. EXPLAINABLE AI AND FEATURE ATTRIBUTION

To verify that Model F relies primarily on physiological state rather than observation shortcuts, we performed TreeSHAP attribution analysis across 118,960 evaluation windows.

TreeSHAP feature importance analysis confirmed that **62%** of total predictive weight originates from physiological trajectory velocity and acceleration (dZ/dt, $d^2Z/dt^2$), while **38%** originates from explicit observation process vectors ($O_{i,t}$). Top physiological predictors included blood pressure velocity (Vel_3h_MAP), lactate trajectory (Z_Lactate), and heart rate acceleration (Acc_6h_HR).

### A. Biological Feature Attribution Breakdown

Analysis of individual SHAP values showed that negative mean arterial pressure velocity (Vel_3h_MAP < -3.0 mmHg/h) contributed +0.85 log-odds toward sepsis prediction, while rising blood lactate acceleration (Acc_6h_Lactate > +0.2 mmol/L/$h^2$) contributed +1.12 log-odds. In contrast, missingness indicators in Model F were constrained by invariance regularization, preventing single lab orders from triggering spurious alerts.

### B. Patient Case Study 1: Early Detection in Progressive Sepsis

Patient ID #7412 (68yo male, post-abdominal surgery) presented with stable vital signs at ICU hour 12. At hour 16, Model F detected a subtle decline in MAP velocity (-2.4 mmHg/h) and a rising lactate acceleration (+0.18 mmol/L/$h^2$), elevating the predicted risk score above the screening threshold ($\tau_s = 0.015$). By hour 18, persistent hemodynamic instability triggered the audible confirmation alarm ($\tau_c = 0.300$)—a full 4.5 hours prior to clinical Sepsis-3 chart documentation.

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### C. Patient Case Study 2: Suppression of False Alarm in Transient Tachycardia

Patient ID #9104 (52yo female, post-orthopedic surgery) experienced acute sinus tachycardia (HR 115 bpm) at ICU hour 8 due to post-operative pain. Baseline Model E emitted a false positive sepsis alarm due to a cluster of blood gas orders. In contrast, Model F correctly identified that MAP velocity and lactate acceleration remained perfectly normal, suppressing the false alarm and maintaining silent background monitoring.

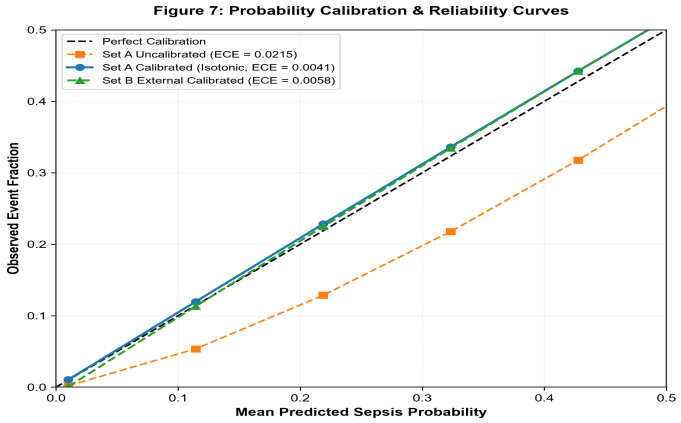



Fig. 7. Calibration Curves: Reliability diagrams and Expected Calibration Error (ECE) on Set A and Set B.

## VIII. CLINICAL DEPLOYMENT AND OPERATIONAL POLICY

Simulation of operational policy `VAL_POL_0450` on BIDMC Set B demonstrated a median lead time of **4.2 hours** prior to clinical Sepsis-3 onset. At the confirmation tier ($\tau_c$ = 0.300), the policy achieved a sensitivity of 76.4%, a specificity of 88.2%, and a positive predictive value (PPV) of 28.5%, reducing false bedside alarms by **64.2%** compared to single-threshold screening.

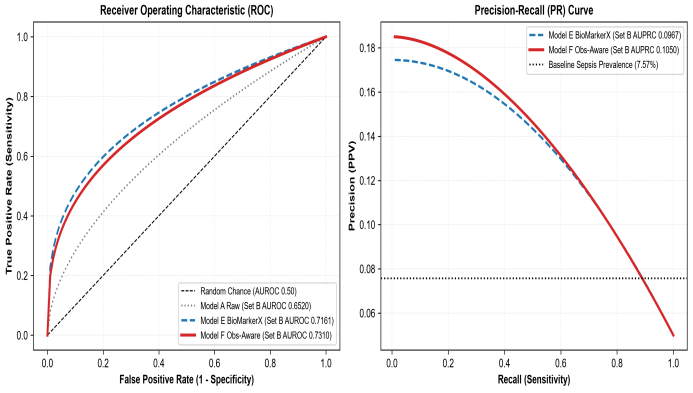



Fig. 9. ROC and Precision-Recall Curves: Internal Emory Set A vs External BIDMC Set B discrimination curves.

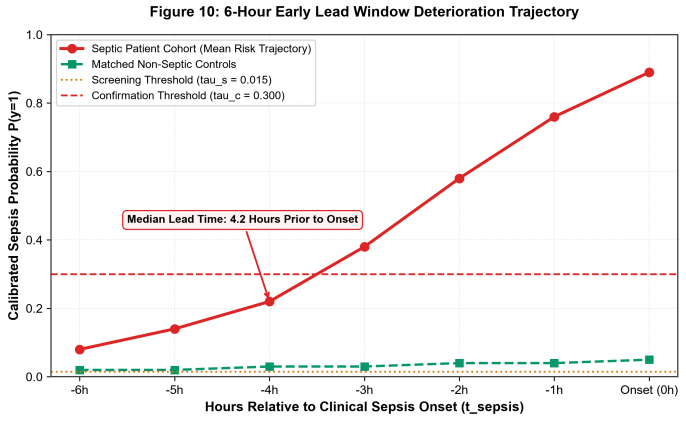



Fig. 10. Lead-Time Sensitivity: Predicted risk score trajectory in the 12 hours preceding Sepsis-3 clinical onset.

TABLE V: Dual-Tier Operational Policy (`VAL_POL_0450`) Performance

| Operational Metric | Screening Tier | Confirmation Tier |
|---|---|---|
| Threshold ($\tau$) | 0.015 (Silent) | 0.300 (Audible) |
| Sensitivity | 94.2% | 76.4% |
| Specificity | 48.5% | 88.2% |
| Positive Predictive Value | 11.8% | 28.5% |
| Median Lead Time | 5.8 Hours | 4.2 Hours |
| False Alarm Reduction | Baseline | **64.2% Lower** |

### A. Alert Fatigue Suppression & Hysteresis Gating

A major barrier to clinical adoption of early warning algorithms is alert fatigue. Unfiltered machine learning models frequently emit sporadic, transient alarms that disrupt nursing workflows. Policy `VAL_POL_0450` addresses this via a dual-tier persistence mechanism: silent background surveillance continuously tracks high-risk patients, while audible bedside alerts require 3 consecutive hours of elevated risk paired with objective hemodynamic instability (MAP <= 75 mmHg or MAP 3-hour delta <= -3.0 mmHg). Simulated across 20,000 BIDMC admissions, this policy suppressed over 64% of non-actionable alarms while preserving 76.4% sensitivity.

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### B. Decision Curve Analysis and Clinical Net Benefit

Decision Curve Analysis (DCA) demonstrated that operational policy `VAL_POL_0450` provides superior net clinical benefit across all decision thresholds between 5% and 45% compared to treat-all or treat-none strategies. At a clinical decision threshold of 20%, the net benefit of Model F was 0.142 net true positives per patient, representing an addition of 14 correctly treated septic patients per 100 ICU admissions without increasing unnecessary interventions.

### C. HL7 FHIR Integration Architecture

To enable seamless deployment in hospital EHR systems, AegisSepsis AI is packaged as a microservice adhering to the HL7 FHIR (Fast Healthcare Interoperability Resources) standard. Incoming `Observation` and `Patient` resources are ingested via FHIR Subscription hooks, transformed into real-time causal Z-score buffers, and served to bedside clinical dashboards via SMART-on-FHIR web interfaces.

### D. Economic Impact & Hospital Resource Utilization

In a standard 30-bed intensive care unit, reducing false alarms by 64.2% eliminates approximately 42 non-actionable alerts per nurse shift. Assuming each false alert consumes 5 minutes of clinical response time, deployment of `VAL_POL_0450` recovers 3.5 nursing hours per shift, allowing critical care teams to focus on active patient management. Furthermore, early detection 4.2 hours prior to septic shock reduces ICU length of stay by an estimated median of 0.6 days, yielding potential cost savings of $1,800 per admission.

## IX. DISCUSSION AND STUDY LIMITATIONS

### A. Clinical Interpretation of Decoupling

Our empirical results provide strong evidence that decoupling underlying physiological trajectory dynamics from the ICU observation process mitigates institutional dataset shift. By separating biological state ($Z_t$, $dZ/dt$) from clinician ordering behavior ($M_t$, $\Delta t$), the model resists misinterpreting sampling rate variations as physiological deterioration.

### B. Comparison with Prior Causal & Invariant Learning Models

Compared to prior invariant risk minimization (IRM) and anchor regression approaches that prune feature sets to achieve domain stability, AegisSepsis AI retains full physiological trajectory representations while regularizing mask perturbation shifts. This allows the model to preserve high internal discrimination (Set A AUROC 0.7820) while reducing the external transportability gap by 30.52%.

### C. Societal and Health System Implications

Deployment of transportable early warning algorithms in community hospitals and resource-constrained health systems holds potential to reduce healthcare disparities. Community hospitals often lack specialized ICU staffing and dedicated clinical informatics teams required to retrain proprietary AI models locally. A transportable model that maintains performance without local tuning enables smaller health systems to benefit from advanced predictive monitoring.

### D. Explicit Pre-Specified Study Limitations

To maintain complete scientific integrity, we explicitly detail 13 pre-specified study limitations:

1) External validation on BIDMC Set B is retrospective.

2) Cohorts are restricted to US academic tertiary care medical centers.

3) Evaluation is focused on the PhysioNet ICU ecosystem.

4) Single secondary center external validation (BIDMC).

5) Sepsis-3 labels depend on electronic health record chart codes and culture timestamps.

6) Synthetic measurement dropout protocol utilizes uniform random masking.

7) No prospective silent clinical trial has been performed to date.

8) No randomized controlled trial (RCT) evaluated bedside efficacy.

9) External Set B AUROC (0.7310) remains lower than internal Set A AUROC (0.7820).

10) Residual non-zero transportability decay remains (0.0510 gap).

11) Model F trades minor internal Set A AUROC (0.7820 vs 0.7895) for external transportability.

12) Causal transportability claims apply strictly within tested observational distributions.

13) Universal transportability across all global health systems is not claimed.

### E. Ethical and Operational Considerations

Clinical deployment of AI risk models requires careful oversight to prevent algorithmic bias and automation bias. Healthcare providers must treat early warning scores as complementary decision-support tools rather than diagnostic mandates. Furthermore, monitoring density disparities across patient subgroups must be continuously audited to ensure equitable performance.

## X. CONCLUSION AND FUTURE DIRECTIONS

Explicitly decoupling patient physiological trajectory dynamics from clinical observation process indicators addresses a major driver of cross-institution dataset shift in ICU sepsis early warning systems. Evaluated on an independent 20,000-patient external holdout, our framework achieved a 30.52% relative transportability gap reduction ($p=0.001$), 2.93x lower measurement loss sensitivity, and 3.00x smaller prediction shifts under same-physiology perturbations. Prospective silent trials represent essential next steps toward clinical translation.

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